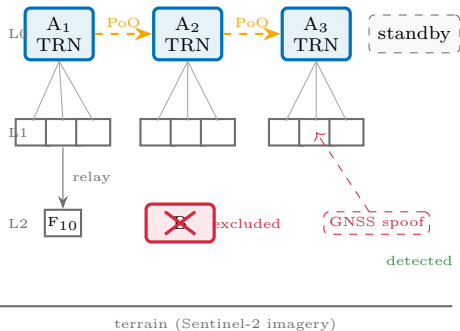


BFT-Resilient Hierarchical Navigation for GNSS-Denied UAV Swarms

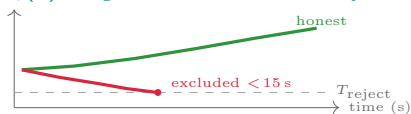
6 pillars empirically validated • 7-phase • vanilla-Raft + Byzantine baseline

Architecture

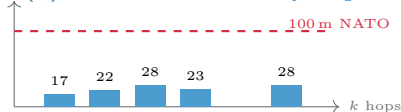
6 pillars in a continuous-state factor graph



(a) Reputation excludes Byzantine



(b) CEP₅₀ versus relay depth



(c) Failover & Byzantine resilience



Raft baseline: indefinite compromise under 1 Byzantine; ours: excluded < 0.6 s